

Prevalence and Mortality due to Outbreak of Novel Coronavirus Disease (COVID-19) in β -Thalassemias: The Nationwide Iranian Experience --Manuscript Draft--

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Abstract:	Background The data on Coronavirus Disease 2019 (COVID-19) in patients with β-thalassemias are scarce. Our aim was to determine the prevalence and mortality rate of patients with β-thalassemias compared to the infected general population in Iran, where it is a hot spot for both the prevalence of β-thalassemias and COVID-19 disease. Methods A multicenter, retrospective cross-sectional study was conducted across all comprehensive thalassemia centers in Iran, from January to April 2020. All suspected and confirmed COVID-19 cases, from a total of 15,950 transfusion-dependent (TDT) and 2,400 non-transfusion dependent thalassemia patients (NTDT) registered by the Iranian Ministry of Health were evaluated. Epidemiological, clinical, imaging and outcome data were extracted from their medical records, using a standardized data collection form. Findings Fifteen confirmed cases of COVID-19 (12 in TDT and 3 in NTDT) were detected from a total of 18,350 β-thalassemia patients. Concomitantly, 8 symptomatic patients with β-thalassemia (6 TDT and 2 NTDT) who were suspected of coronavirus disease, without performing test were also analyzed. The mean age of confirmed and suspected patients was 36.1 ± 12.1 (range: 22-66) and 39.6 ± 9.03 (range: 30-54), respectively. Eleven confirmed patients had mild to moderate symptoms and recovered, and four severe cases, with at least one comorbidity, died. The prevalence of coronavirus disease in the general Iranian population was 11.01 and in patients with β-thalassemias was 8.17 per 10,000 population ($P:0.246$). The mortality rate of coronavirus disease was significantly higher in patients with β-thalassemias (26.6%) compared to the general population (6.34%) ($P: 0.001$). Interpretation To the best of our knowledge, this is the largest nationwide retrospective study among thalassemic patients with COVID-19 who have experienced a definite outcome. Our findings provide further objective evidence to consider a comprehensive risk assessment and prognosis among thalassemic patients with COVID-19. Early vigilant monitoring along with high quality supportive care is needed albeit more detailed information should be collected in patients with thalassemia in future studies.

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Prevalence and Mortality due to Outbreak of Novel Coronavirus Disease (COVID-19) in β -Thalassemias: The Nationwide Iranian Experience

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Summary

Background The data on Coronavirus Disease 2019 (COVID-19) in patients with β -thalassemias are scarce. Our aim was to determine the prevalence and mortality rate of patients with β -thalassemias compared to the infected general population in Iran, where it is a hot spot for both the prevalence of β -thalassemias and COVID-19 disease.

Methods A multicenter, retrospective cross-sectional study was conducted across all comprehensive thalassemia centers in Iran, from January to April 2020. All suspected and confirmed COVID-19 cases, from a total of 15,950 transfusion-dependent (TDT) and 2,400 non-transfusion dependent thalassemia patients (NTDT) registered by the Iranian Ministry of Health were evaluated. Epidemiological, clinical, imaging and outcome data were extracted from their medical records, using a standardized data collection form.

Findings Fifteen confirmed cases of COVID-19 (12 in TDT and 3 in NTDT) were detected from a total of 18,350 β -thalassemia patients. Concomitantly, 8 symptomatic patients with β -thalassemia (6 TDT and 2 NTDT) who were suspected of coronavirus disease, without performing test were also analyzed. The mean age of confirmed and suspected patients was 36.1 ± 12.1 (range: 22-66) and 39.6 ± 9.03 (range: 30-54), respectively. Eleven confirmed patients had mild to moderate symptoms and recovered, and four severe cases, with at least one comorbidity, died. The prevalence of coronavirus disease in the general Iranian population was 11.01 and in patients with β -thalassemias was 8.17 per 10,000 population ($P:0.246$). The mortality rate of coronavirus disease was significantly higher in patients with β -thalassemias (26.6%) compared to the general population (6.34%) ($P: 0.001$).

Interpretation To the best of our knowledge, this is the largest nationwide retrospective study among thalassemic patients with COVID-19 who have experienced a definite outcome. Our findings provide further objective evidence to consider a comprehensive risk assessment and prognosis among thalassemic patients with COVID-19. Early vigilant monitoring along with high quality supportive care is needed albeit more detailed information should be collected in patients with thalassemia in future studies.

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Research in context

Evidence before this study

The data on Coronavirus Disease 2019 (COVID-19) in patients with β -thalassemias are scarce. A small cohort study from Northern Italy, that was the epicentre for coronavirus COVID-19 in Europe, experienced relatively mild to moderate COVID-19 disease compared to general population and all infected thalassemia patients recovered.

Added value of this study

This is the first nationwide COVID-19 study, to date, with a major focus on the epidemiological characteristics and outcome in patients with thalassemias and COVID-19. All suspected and confirmed COVID-19 cases, from a total of 15,950 transfusion-dependent (TDT) and 2,400 non-transfusion dependent thalassemia patients (NTDT) registered by the Iranian Ministry of Health were evaluated. Fifteen confirmed cases of COVID-19 (12 in TDT and 3 NTDT) were detected from a total of 18,350 β -thalassemia patients. Concomitantly, eight symptomatic β -thalassemia patients (6 TDT and 2 NTDT) with suspected coronavirus disease were also collected. Seventeen confirmed and suspected patients (73.9%) had mild to moderate symptoms and recovered, while 6 patients died (26.1%), 4 (26.6%) in the confirmed group of COVID-19 cases and 2 (25%) in the suspected group of patients. Four out of twelve TDT patients with confirmed COVID-19 had insulin-dependent diabetes. One of them, a female patient, died at the age of 30 year.

Implications of all the available evidence

The prevalence of coronavirus disease in the general Iranian population was 11.01 and in patients with β -thalassemias was 8.17 per 10,000 population (P: not statistically significant). Mortality rate of coronavirus disease was significantly higher in patients with β -thalassemias (26.6%) compared to the general population (6.34%) (P: 0.001). These findings provide further objective evidence to take into account in the comprehensive risk assessment and prognosis among thalassemic patients with COVID-19.

Introduction

In late December 2019, an ongoing outbreak of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection that was termed Coronavirus Disease 2019 (COVID-19), was reported in Wuhan, Hubei Province, China. During the following weeks, infections spread across China and other countries around the world. The novel coronavirus outbreak that can cause diseases ranging from the common cold to more severe and even fatal respiratory infections¹.

A total of 3,018,681 have been reported globally and 92,584 confirmed cases have been documented in Iran, until April 29th, 2020. The death toll from the COVID-19 outbreak was 207,973 worldwide, and 5,877 in Iran ². Fever and cough were the most prevalent symptoms at disease onset in both deceased patients and recovered patients. Other prevalent symptoms at onset of illness in deceased patients included fatigue, dyspnoea, chest tightness, and sputum production; fewer common symptoms included anorexia, diarrhoea, and myalgia. Dyspnoea and chest tightness were much more common in deceased patients than in recovered patients ³.

The diagnosis of COVID-19 must be confirmed by reverse-transcription polymerase chain reaction (RT-PCR) or gene sequencing for respiratory or blood specimens, but at the presentation some confirmed cases could have negative results ⁴. Therefore, the putative role of chest computed tomography (CT) has been considered, as a primary tool for the current COVID-19 detection, in epidemic areas ⁵.

β -thalassemias are a group of hereditary autosomal recessive anemias caused by either reduced or complete absence of production of β -globin chains of the hemoglobin tetramer. Thalassemia syndromes are one of the most prevalent genetic anemias affecting 7% of the world population, and are evolving globally due to migration^{6, 7}. Three clinical and hematological conditions are recognized based mainly on the severity of clinical features of disease i.e., the β -thalassemia carrier state, non-transfusion-dependent and transfusion-dependent thalassemia (NTDT and TDT) ⁸.

Patients with NTDT have milder clinical manifestations and hemoglobin levels above 7-8 g/dl, at least for the first 1-2 years of life, whereas TDT have severe anemia usually within the first two years of life, requiring regular red blood cell (RBC) transfusions^{6, 8}. In Iran, the homozygote β -thalassaemia patients are about 20,000 and 3,750,000 are the carriers ⁹.

There is no evidence that β -thalassemia carriers are at an increased risk of becoming infected by SARS-CoV-2. However, in TDT and NTDT patients, anemia and iron-overload can involve the heart, lungs (pulmonary hypertension), liver, endocrine glands and even immune system, and can expose them to a potentially higher risk of complications due to COVID-19, compared to normal population, especially in older patients age group ¹⁰.

Limited data about frequency and outcomes of infected COVID-19 patients with thalassemia are currently available in the literature. In this study, the primary aim was to determine the prevalence, severity and mortality rate of COVID-19 in patients with TDT and NTDT living in Iran. Moreover, we evaluated the associated risk factors in this vulnerable group of patients.

Methods

Study participants and data collection

This is a multicenter, retrospective, cross-sectional study that was obtained across all comprehensive thalassemia centers in Iran, from January to April 29th, 2020. All suspected and confirmed COVID-19 cases from a total of 15,950 TDT and 2,400 NTDT patients registered by the Iranian Ministry of Health were evaluated.

Epidemiological, demographics, clinical characteristics (type of thalassemia, splenectomy status, associated complications), laboratory, imaging, and outcome data were collected analyzing patients' electronic medical records.

Diagnosis, classification, and management

The coronavirus test was performed based on the Berlin protocol by real-time RT-PCR ¹¹.

Clinical and imaging manifestations for diagnosis of suspected cases were: fever, shortness of breath, fatigue, dry cough, and chest computed tomography (CT) scan imaging suggestive of SARS-CoV-2 pneumonia infection.

All patients were classified as mild, moderate, severe, and critical ill according to the guidance issued by the National Health Commission of China ¹².

Mild patients present mild symptoms without radiographic features. Moderate patients present with fever, respiratory symptoms, and radiographic features.

Severe patients meet one of three criteria: (a) dyspnea, RR greater than 30 times/min, (b) oxygen saturation less than 93% in ambient air, and (c) PaO₂/FiO₂ less than 300 mmHg. Critical patients meet one of three criteria: (a) respiratory failure, (b) septic shock, and (c) multiple organ failure.

According to the Iranian Ministry of Health protocol, confirmed cases and suspected symptomatic patients (fever, shortness of breath with respiratory rate > 30/minutes and hypoxia with SpO₂ < 93% or with chest CT scan positive for COVID-19) were referred to a dedicated COVID-19 hospital for specific follow-up and treatment. Hydroxychloroquine combined with lopinavir/ritonavir or atazanavir were the suggested treatment regimens (https://corona.ir/media/upload/1399/02/_chart_tshkhys_w_drman_srpayy_w_bstry_kwwyd19_3_ardybhsht1-5ea4010ce224b.pdf).

The study protocol was approved by the Ethical Committee of Shiraz University of Medical Sciences (Code:1399.228), following the Helsinki Declaration and the good clinical practices. The patients or their legal guardians signed a consent form to protect the confidentiality and their identity.

Statistical analysis

Data were analyzed using Statistical Package for the Social Sciences (SPSS) software version 23 (SPSS Inc., Chicago, Illinois, USA).

Descriptive information was presented mean, standard deviation, median, range, frequency, and percentage. A comparison of qualitative and quantitative variables was done by Fisher's exact test and Mann-Whitney test respectively between the two groups of confirmed COVID-19 cases with the outcomes of recovery or death. COVID-19 prevalence and mortality rate in patients with TDT and NTDT were calculated in confirmed cases by PCR and compared, by the Chi-square test, with those reported in the general Iranian population using MedCalc software. P value ≤ 0.05 was considered statistically significant.

Results

During the study period, 15 confirmed cases of COVID-19 (12 TDT and 3 NTDT) were registered, from a total number of 18,350 Iranian β-thalassemia patients.

Demographics, clinical characteristics, and laboratory data of patients are presented in table 1. Concomitantly, 8 symptomatic β -thalassemia patients (6 TDT and 2 NTDT), who were suspected for coronavirus disease, without PCR confirmation, were also collected. Clinical characteristics and laboratory data of all confirmed and suspected COVID-19 cases are summarized in table 2.

The mean age in confirmed and suspected COVID-19 patients with TDT and NTDT was 36.1 ± 12.1 years (range: 22-66 years) and 39.6 ± 9.03 years (range: 30-54 years), respectively. Sixty one percent of all confirmed and suspected patients with thalassemias were males. The mean hemoglobin level was 8.9 ± 0.68 g/dl in confirmed cases and 8.4 ± 1.44 g/dl in suspected COVID-19 patients. Serum ferritin levels were above 2,000 ng/ml in 21.4% of the RT-PCR positive and 42.9% in suspected not-tested group.

More than 60% in each confirmed and suspected groups had at least one comorbidity consisting of insulin-dependent diabetes, heart disease, pulmonary artery hypertension, hypertension due to kidney function disease, osteoporosis, history of hepatitis C virus (HCV) positivity, liver disease, and asthma (table 2).

Six patients died (26.1%), four (26.6%) in the confirmed group of COVID-19 cases and 2 (25%) in the suspected group of patients.

In order to determine the influence of associated factors to disease severity, we compared the clinical, therapeutic and laboratory data including sex, age, thalassemia type, splenectomy, hemoglobin level, serum ferritin level, heart or liver iron overload, white blood cells and platelet counts, hydroxyurea treatment and presence of comorbidities including major underlying diseases, in RT-PCR positive β -thalassemia patients who recovered versus those who deceased (table 3).

The two groups were comparable in nearly all variables except for the thalassemia type. In the group of patients presenting recovery, the frequency of TDT patients was significantly higher than the deceased group (100% vs. 25%; $P: 0.009$). Moreover, major underlying diseases including diabetes, heart disease, hypertension due to kidney function disease, pulmonary artery hypertension, and asthma showed a borderline significant higher frequency in the deceased group compared to the recovered group (100% vs. 36.4%; $P: 0.051$). All three NTDT patients as well as one TDT patient who died had at least one major underlying disease.

Then we compared all above-mentioned variables between TDT and NTDT patients. The two groups were similar regarding all variables ($P > 0.05$, data was not shown) except hemoglobin which was significantly higher in NTDT patients [median 9.8 g/dl (range: 9.2-9.9) vs. 8.6 g/dl (range: 7.9-9.7), respectively ($p: 0.030$)]. Frequency of patients with at least one major underlying disease was 41.7% in TDT and 100% in NTDT ($P: 0.123$).

The COVID-19 prevalence and mortality rate, up to April 29th, 2020 was also calculated in patients with β -thalassemias and in patients of general Iranian population (table 4).

The prevalence of COVID-19 was 11.01/10,000 in general population compared to 8.17/10,000 in patients with β -thalassemia. The difference was not statistically significant ($P: 0.246$). On the other hand, at the same date, the mortality rate was significantly higher in patients with β -thalassemia (26.6%) compared to the general population (6.34%) ($P: 0.001$).

In our survey, the source of infection has been identified only in a few of cases, two from their families, one from travel to Qom where was the first point of epidemic in Iran and one was working at hospital.

Discussion

Coronavirus disease 2019 (COVID-19) is a potentially severe acute respiratory infection caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). It is a zoonotic disease with person-to-person transmission through droplet or contact transmission¹³. It is uncertain how long an individual with COVID-19 remains infectious. The viral shedding duration seems to vary according to the disease severity. It has been found that around 90% of patients with milder symptoms had a negative viral RNA test on nasopharyngeal swabs by day 10 post-onset, while the test remained positive for a longer time in all severe cases¹⁴.

In a large Chinese report, including 72,314 patients, only 2% of those infected were younger than 20 years old, the majority of cases (89.8%) were between the age of 30 to 79 years old. The overall case mortality rate was 2.3% but this rate increased proportionally with age reaching 8% in those aged 70- 79 years and 14.8% in those > 80 years old¹⁵. Patients with COVID-19 who required intensive care (IC) treatment were more likely to have diabetes (22.2% v 5.9%) compared to those who did not require IC admission¹⁶.

According to COVID-19 webinar presented by European Hematology Association, patients with TDT and COVID-19 have been reported in Cyprus, Italy, Pakistan, Iran, UK, France, Indonesia, and Turkey ([https://ehaweb.org/covid-19/webinars.session 3; April 16th, 2020](https://ehaweb.org/covid-19/webinars.session%203%20April%2016th%202020)). Most of them presented mild- moderate respiratory symptoms (46 out of 51) and 3 out of 5 hospitalized patients died.

A small cohort study from Northern Italy, that was the epicentre for coronavirus COVID-19 in Europe, experienced relatively mild to moderate COVID-19 disease (11 cases:10 with TDT and 1 with NTDT) compared to the general population and all infected thalassemia patients recovered¹⁷. One patient with severe symptoms required ventilation support with continuous positive airway pressure (CPAP). Their mean age was 44 ± 11 years (range 31-61 years) and 55% (6/11) were females. All patients had associated comorbidities and 70% of them were splenectomized. The likely source of infection was found in 64% of cases ¹⁷.

Our survey is the first nationwide investigation that systematically evaluated the prevalence of COVID-19, the presence of comorbidities and the prognosis in patients with TDT and NTDT living in Iran.

The prevalence of COVID-19, in patients with thalassemias, was lower than the overall disease prevalence reported in the Iranian population (8.17 vs.11.01/10,000 population) but the difference was not statistically significant (P: 0.246).

Moreover, most of our confirmed COVID-19 patients showed mild to moderate disease (73.3 %, 11 out of 15 cases), as reported in Italy ¹⁷, but 4 out of 15 confirmed COVID-19 cases (26.6%) developed severe symptoms contributing to patients' mortality (1 TDT and 3 NTDT, aged 30-60 years; mean: 38.75 ± 14.36 , median: 32.5; 2 males) while no deaths were reported in Italian patients with thalassemias ¹⁷.

The reason why the prevalence of COVID-19 disease in β -thalassemia patients is less than general population, despite underlying diseases is not known yet. The hypothesis of attacking Coronavirus to β -globin chain of hemoglobin as β -thalassemia patients have a decreased β -globin production that leads to less attack by Coronavirus¹⁸, ineffective erythropoiesis ^{6, 8} and increased clearance of infected red blood cells by the virus might be possible mechanisms, but a substantial amount of evidence is still needed to support the proposed mechanisms.

The mortality rate in COVID- 19 RT-PCR positive thalassemic patients was significantly higher (26.6 %) compared to the general Iranian population (6.34%) (P:0.001) which may be attributed to the presence of additional risk factors in patients with thalassemias¹⁰. In patients with TDT, suboptimal treatment or poor compliance to iron chelation therapy is associated with multiple organ damage,

including insulin-dependent diabetes, heart failure or arrhythmia and liver disease, whereas hypercoagulable state and pulmonary hypertension are more common complications in NTDT patients^{8,10}.

More than 30 % of both RT-PCR confirmed and suspected COVID - 19 cases had moderate to severe heart and liver iron overload which underscore the importance of adherence to iron chelation regiment in order to reduce the risk of comorbidities and severity of SARS-CoV-2 infection. Four out of twelve TDT patients with RT-PCR confirmed COVID-19 had insulin-dependent diabetes. One of them, a young female patient, died at the age of 30 year (table 1).

Diabetes may increase the risk of viral infection because of impaired innate immunity due to impaired macrophage and lymphocytes function which also increases the speed of progression to septic shock and multiple organ failure leading to poor outcomes¹⁹.

Therefore, it is recommended that patients with diabetes maintain a good glycemic control, because it might help reduce the risk of infection itself and may also modulate the severity of the clinical expression of the disease¹⁰. Moreover, it is advised to ensure adequate stock of medications and supplies for monitoring blood glucose during the period of home confinement. Social distancing as well as home confinement must be adopted and crowd's hospital waiting rooms should be avoided²⁰.

Our data demonstrated a borderline significantly higher frequency of major underlying diseases in the deceased group compared to the recovery group. Moreover, a non-significant higher frequency of major underlying diseases in NTDT compared to TDT patients (100% vs 41.7%) despite a significantly higher mortality rate in NTDT patients was reported in our study. These findings can be due to the small number of patients that may have reduced the statistical power and indicates the necessity of a larger sample size to evaluate the differences between COVID-19 positive patients with TDT and NTDT.

Eighty per cent of our patients with confirmed COVID - 19 were splenectomized but splenectomy was not significantly associated with the fatal outcome (table 3).

There are not enough data showing a relationship between blood groups and COVID-19 although one observational study from China reported an association with blood groups. Subjects with A blood group were at higher risk whilst subjects

with blood group O were associated with a lower risk for SARS-CoV-2 infection²¹.

Our study also shows a low frequency of blood group O among confirmed COVID-19 cases (two out of 15), although we cannot conclude that blood group O is associated with lower risk for COVID-19 disease due to small number of documented patients.

No cases of COVID-19 in the pediatric and adolescent thalassemia population were observed. Our result is similar to fewer children's involvement by COVID-19 in the general population²². Nevertheless, special attention should be devoted to thalassemic children with severe iron overload because they could be more vulnerable to SARS-CoV-2 infection.

Sixty one percent of all confirmed and suspected patients with thalassemias were males, suggesting the presence of a possible sex-dependent susceptibility.

Channappanavar et al.²³ have shown that male mice are more susceptible to SARS-CoV-2 infection than age-matched females. Furthermore, ovariectomy or treatment of female mice with an estrogen receptor antagonist increased the mortality rate; therefore, suggesting a protective effect for the estrogen receptor signalling pathway in SARS-CoV-2 infected mice²³. In this context, the analysis of a possible hormonal-dependency of the expression and/or activity of ACE2 in the various tissues is of relevance to understand the pathologic mechanisms behind the epidemiologic findings.

There is currently not available vaccine or specific antiviral treatment for Coronavirus. Based on local country protocol, our patients received hydroxy chloroquine alone or combined with lopinavir/ritonavir or atazanavir, but the effectiveness of such regimen is still an open question.

During the Coronavirus outbreak in Iran, most of thalassemia treatment centers were guided by the Ministry of Health to implement measures including: emergency phone calls in case of developing symptoms like fever, cough or respiratory difficulty, wearing masks, gloves and shield by medical staff, avoid being exposed and leaving six feet distance between beds for blood transfusion, and referring symptomatic patients to the dedicated COVID-19 hospitals for testing and hospitalization, if needed.

Moreover, feasibilities for patients to get medications including iron chelation medicines and isolated rooms for blood transfusions, in suspected or confirmed cases, to decrease the rate of infection were recommended, although these recommendations were not very easy to implement in some areas of Iran, due to limited facilities.

The present study has several strengths and limitations. First, this is the first nationwide study, to date, with a major focus on the epidemiological characteristics, the prevalence and outcome in patients with thalassemias and COVID-19 compared to infected normal population contracted with COVID-19. Second, we included both confirmed and suspected COVID-19 cases, to have a more comprehensive picture of thalassemic patients with COVID-19 disease. About limitations, we were unable to assess some laboratory characteristics of thalassemic patients positive for COVID-19 and adequate information on patient's exposure history were not fully available, and thus, the incubation period was not examined in our study.

In conclusion, our study shows a comparable prevalence of COVID-19 in thalassemia patients versus general Iranian population. Most of β -thalassemia patients developed mild to moderate disease and recovered but having multiple comorbidities can predispose them to have a severe disease and a significant higher risk of mortality rate in thalassemia patients compared to infected general population with COVID-19. The minimum age for disease occurrence and death was 22 and 30 years in the confirmed COVID-19 patients with TDT, and 30 and 34 years in suspected cases. These findings provide further objective evidence to take into account in the comprehensive risk assessment and prognosis among thalassemic patients with COVID-19.

We are facing many unknowns, since the epidemic of COVID-19 is on-going and rapidly evolving, more detailed information should be collected in patients with thalassemia in future studies.

Contributors

MK made substantial contributions to the study concept, design the protocol, and in drafting the manuscript. SH performed the statistical analysis and participated in drafting the manuscript. AA, ZZ, MA, AS, and AB were responsible for data collection and confirmation. TZ participated in drafting the manuscript.

VDS participated to the preparation of protocol survey and in drafting the manuscript and revising it.

Declaration of interest

The authors report no conflicts of interest.

Dedication and acknowledgements

We express our appreciation to thalassemia patients and sincere condolences to the families who have lost thalassemic patients with COVID-19 disease. We would like to express our appreciation to Iranian Thalassemia Society, all Iranian thalassemia Centers, doctors, and nurses for the very kind cooperation and help in data collection. We also thank Herand Zargarian for English editing.

No.	Age (yrs)	Sex	Disease type	Blood group	Spleen	Comorbidities	Hb level (g/dl)	Serum ferritin (ng/ml)	EF (%)	PAP (mmHg)	Outcome
1	30	M	NTDT	A-	Yes	cholangitis, PAH, and reduced EF	9.9	311	45	40	dead
2	22	M	TDT	B+	No	none	7.9	329	60	15	recovered
3	36	M	TDT	A+	yes	osteoporosis, history of HCV positivity	9.0	901	60	23	recovered
4	32	F	TDT	B+	yes	diabetes, PAH, heart disease	9.6	2338	55	32	recovered
5	38	F	TDT	AB+	yes	history of HCV positivity	8.3	2000	60	21	recovered
6	31	F	TDT	B+	yes	diabetes	8.0	830	55	29	recovered
7	22	M	TDT	B+	yes	none	8.5	253	60	28	recovered
8	38	F	TDT	A+	No	none	9.7	800	55	25	recovered
9	30	F	TDT	B+	No	diabetes, HTN and renal disease	8.2	266	60	28	dead
10	36	M	TDT	B+	yes	history of HCV positivity	9.0	1400	55	28	recovered
11	35	F	NTDT	AB+	yes	osteoporosis, diabetes, PAH, reduced EF	9.8	600	45	35	dead
12	60	M	NTDT	B+	yes	PAH, reduced EF, cholecystectomy	9.2	1962	45	80	dead
13	38	M	TDT	O+	yes	osteoporosis, history of HCV positivity	8.5	5460	55	28	recovered
14	28	F	TDT	O+	yes	asthma	9.6	8200	55	28	recovered
15	66	F	TDT	A+	yes	diabetes, osteoporosis	8.7	225	55	25	recovered

Legend = TDT: transfusion-dependent β -thalassemia, NTDT: non-transfusion-dependent β -thalassemia, EF: ejection fraction, PAP: pulmonary artery pressure, HTN: hypertension, Spleen: splenectomy, PAH: pulmonary artery hypertension, PAH was defined as PAP > 30 mmHg; EF: ejection fraction, low EF was defined as EF < 50%

Table 1. Data of confirmed COVID-19 in patients with β -thalassemias

Parameters	COVID- 19 (RT-PCR positive; n=15)	Suspected COVID - 19 (n=8)
Age (year), Mean $\pm$ SD, min-max	36.1 $\pm$ 12.1, 22-66	39.6 $\pm$ 9.03, 30-54
Sex (m/f)	7/8	7/1
Thalassemia type (N %)	=	=
TDT	12 (80)	6 (75)
NTDT	3 (20)	2 (25)
Splenectomy (yes) (N %)	12 (80)	4 (57.1)
Serum ferritin (ng/ml), Mean $\pm$ SD	1,725 $\pm$ 2,245	2,847 $\pm$ 3,189
Median (min-max)	830 (225-8,200)	2,000 (545-9,556)
(N %)	=	=
≤ 2000	11 (78.6)	4 (57.1)
> 2000	3 (21.4)	3 (42.9)
Hemoglobin (g/dl), Mean $\pm$ SD	8.9 $\pm$ 0.68	8.4 $\pm$ 1.44
WBC (per μ L), Mean $\pm$ SD	15,885 $\pm$ 11,579	22,914 $\pm$ 15,830
Platelet count (per μ L), Mean $\pm$ SD	530,820 $\pm$ 289,787	514,428 $\pm$ 213,550
Heart or liver iron overload (moderate and severe) (N %)	5 (33.3)	3 (37.5)
Hydroxyurea (yes) (N %)	3 (20)	4 (50)
Comorbidities (yes) (N %)	12 (80%)	5 (62.5)
Major underlying disease*	8 (53.3)	4 (50)
Symptoms (N %)	=	=
Fever	10 (66.7)	3 (37.5)
Cough	9 (60)	2 (25)

Dyspnea	4 (26.6)	2 (25)
Rhinorrhea and sneezing	1 (6.6)	NR
Sinusitis	1 (6.6)	NR
Myalgia	NR	1 (12.5)
Diarrhea	NR	1 (12.5)
Anosmia and hearing loss	1 (6.6)	NR
Weakness in lower extremities	1 (6.6)	NR
Condition	=	=
Recovered	11 (73.3)	6 (75)
Death	4 (26.6)	2 (25)

Legend: RT-PCR: reverse-transcription polymerase chain reaction, TDT: transfusion-dependent β -thalassemia, NTDT: non-transfusion-dependent β -thalassemia, EF: ejection fraction, PAP: pulmonary artery pressure, NR: not reported; * Heart disease, Diabetes, pulmonary artery hypertension, hypertension, and asthma

Table 2. Summary of clinical characteristics and laboratory data of confirmed and suspected COVID-19 cases in patients with β -thalassemias

	Recovered	Dead	P value
	N=11	N=4	
Age (year), Median (min-max)	36 (22-66)	32.5 (30-60)	0.949
Sex (m) (N %)	5 (45.5)	2(50)	> 0.999
Thalassemia type (TDT) (N %)	11 (100)	1(25)	0.009
Splenectomy (yes) (N %)	9 (81.8)	3 (75)	> 0.999
Hemoglobin (g/dl) Median (min-max)	8.7 (7.9-9.7)	9.5 (8.2-9.9)	0.226
Serum ferritin (ng/ml) >2000 (N %)	3 (27.3)	0	> 0.999
White Blood Count (per μL), Median (min-max)	12,300(63-31900)	12,000 (3,775-34,360)	>,0.999
Platelet count (per μL), Median(min-max)	589500 (840-848000)	582500 (147000-965000)	0.808
Hydroxyurea (yes) (N %)	1 (9.1)	2 (50)	0.154
Heart or liver iron overload (moderate and severe) (N %)	4 (36.4)	1 (25)	> 0.999
Comorbidities (N %)	8 (72.7)	4(100)	0.516
Major underlying disease* (N %)	4 (36.4)	4(100)	0.051

* Heart disease, Diabetes, pulmonary artery hypertension, hypertension, and asthma

Table 3. Comparison of demographic and clinical characteristics between confirmed COVID-19 cases in patients with β -thalassemias based on disease outcome

	Total population	Total confirmed cases	Total death	Prevalence per 10,000	P value	Mortality rate (%)	P value
Iran							
General population	84,053,504	92,584	5,877	11.01	0.246	6.34	0.001
β- thalassemia	18,350	15	4	8.17		26.6	

Table 4: Prevalence and mortality rate of COVID-19, up to April 29th, 2020, in general Iranian population and in patients with β-thalassemias

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